

Generate Two 50Hz Waveforms with Variable ON Time (1-2ms) Based on Potentiometers, Reset with Button

Every 20ms, read two potentiometers via ADC and adjust the ON time of two 50Hz PWM waveforms linearly from 1ms (ADC=0) to 2ms (ADC=1023). Start at 1.5ms. A button resets to 1.5ms. Use interrupts for non-blocking operation.

Initialization

1. Register: DDRC a. Value: 0bxxxxxx00 (bits 0 and 1 cleared) b. Configuration(s): i. Configure PC0 as input for ADC0 (potentiometer 1) ii. Configure PC1 as input for ADC1 (potentiometer 2) c. Code: `DDRC &= ~(1 << PC0) | (1 << PC1);`
2. Register: ADCSRA a. Value: 0b10000111 b. Configuration(s): i. Enable ADC (ADEN=1) ii. Set prescaler to /128 (ADPS=111) for ADC clock ~125kHz c. Code: `ADCSRA = (1 << ADEN) | (1 << ADPS2) | (1 << ADPS1) | (1 << ADPS0);`
3. Register: ADMUX a. Value: 0b01000000 (initially for channel 0, changes dynamically) b. Configuration(s): i. Select Vcc as reference (REFS0=1, REFS1=0) ii. Right justify result (ADLAR=0) iii. Select channel (MUX[3:0]=0000 for ADC0, 0001 for ADC1) c. Code: `ADMUX = (1 << REFS0) | channel; // Channel set before each conversion`
4. Register: DDRB a. Value: 0b00000110 b. Configuration(s): i. Configure PB1 as output for OC1A (PWM waveform 1) ii. Configure PB2 as output for OC1B (PWM waveform 2) c. Code: `DDRB |= (1 << PB1) | (1 << PB2);`
5. Register: ICR1 a. Value: 39999 (0x9C3F) b. Configuration(s): i. Set TOP for Fast PWM mode to achieve 20ms period (50Hz) with /8 prescaler (tick=0.5us) c. Code: `ICR1 = 39999;`
6. Register: OCR1A a. Value: 3000 (initial) b. Configuration(s): i. Set compare value for channel A ON time (updated dynamically) c. Code: `OCR1A = 3000;`
7. Register: OCR1B a. Value: 3000 (initial) b. Configuration(s): i. Set compare value for channel B ON time (updated dynamically) c. Code: `OCR1B = 3000;`
8. Register: TCCR1A a. Value: 0b10100010 b. Configuration(s): i. Non-inverting PWM on OC1A (COM1A1=1, COM1A0=0) ii. Non-inverting PWM on OC1B (COM1B1=1, COM1B0=0) iii. Part of mode 14 (WGM11=1, WGM10=0) c. Code: `TCCR1A = (1 << COM1A1) | (1 << COM1B1) | (1 << WGM11);`
9. Register: TCCR1B a. Value: 0b00011010 b. Configuration(s): i. Part of mode 14 (WGM13=1, WGM12=1) ii. Clock prescaler /8 (CS12=0, CS11=1, CS10=0) c. Code: `TCCR1B = (1 << WGM13) | (1 << WGM12) | (1 << CS11);`
10. Register: TIMSK1 a. Value: 0b00000001 b. Configuration(s): i. Enable Timer1 overflow interrupt (TOIE1=1) c. Code: `TIMSK1 = (1 << TOIE1);`
11. Register: DDRD a. Value: 0bxxxxx0xx (bit 2 cleared) b. Configuration(s): i. Configure PD2 as input for push button c. Code: `DDRD &= ~(1 << PD2);`
12. Register: PORTD a. Value: 0bxxxxx1xx (bit 2 set) b. Configuration(s): i. Enable internal pullup on PD2 c. Code: `PORTD |= (1 << PD2);`

13. Register: EICRA a. Value: 0b00000010 b. Configuration(s): i. Falling edge trigger for INT0 (ISC01=1, ISC00=0) c. Code: EICRA = (1 << ISC01);
14. Register: EIMSK a. Value: 0b00000001 b. Configuration(s): i. Enable INT0 interrupt c. Code: EIMSK = (1 << INT0);

Here are the register summaries and functional descriptions, neatly organized by functional area.

Analog-to-Digital Converter (ADC) Setup & Operation

These registers control the ADC's input pins, reference voltage, channel selection, and the conversion process.

Register Name	Function	Detailed Purpose
DDRC	Determines direction of Port C pins (inputs for ADCs).	Must be configured for input for the ADC pins used (typically PC0 and PC1).
ADMUX	Selects ADC reference, alignment, channel.	Bits 0-3 select the ADC channel (0 or 1) and set the reference voltage.
ADCSRA	Controls ADC enable, start, prescaler, flags.	Bit 6 (ADSC) is set to start an ADC conversion, and then polled (wait for ADSC=0) for conversion complete.
ADC	Read 10-bit ADC result.	The result is read from the low byte (ADCL) and high byte (ADCH): $\$ \{ADCL\} \mid \{ADCH\} \ll 8$.

Timer1 (PWM) Setup & Control

These registers configure Timer1 to operate in a Pulse Width Modulation (PWM) mode and define the period and duty cycles.

Register Name	Function	Detailed Purpose
DDRB	Determines direction of Port B pins (outputs for PWM).	Must be configured for output for the PWM output pins (typically PB1/OC1A and PB2/OC1B).
ICR1	Sets TOP value for Timer1 PWM period .	Defines the counter value at which the timer resets, thus setting the PWM frequency/period (20ms in this application).
TCCR1A	Controls Timer1 PWM mode , compare output behavior.	Configures the specific PWM mode and sets the output pins (OC1A and OC1B) behavior (e.g., non-inverting).
TCCR1B	Controls Timer1 clock source , mode.	Selects the prescaler and contributes to setting the overall Timer1 mode.
OCR1A	Sets compare value for Timer1 channel A .	Updates the PWM ON time for waveform 1.
OCR1B	Sets compare value for Timer1 channel B .	Updates the PWM ON time for waveform 2.
TIMSK1	Enables Timer1 interrupts .	Enables the Timer1 Overflow Interrupt .

Interrupt Service Routines (ISRs)

Timer1 Overflow ISR (**TIMER1_OVF_vect**)

Event	Function	Registers/Variables Updated
Runs every 20ms at PWM period end.	Reads both ADCs, computes new ON ticks (2000 to 4000).	OCR1A, OCR1B

External Interrupt ISR (**INT0_vect**)

Event	Function	Registers/Variables Updated
Runs on button press .	Resets ON ticks to 3000 1.5ms and updates PWM outputs immediately.	OCR1A, OCR1B

External Interrupt & Button Setup

These registers configure the external interrupt pin (connected to the button).

Register Name	Function	Detailed Purpose
DDRD	Determines direction of Port D pins (input for button).	The pin connected to the button (INT0 typically PD2) must be configured for input.
PORTD	Sets pullup for Port D input.	Enables the internal pull-up resistor for the button input pin.

EICRA	Controls external interrupt sense .	Configures the interrupt to trigger on a specific edge (e.g., falling edge) or level.
EIMSK	Enables external interrupts .	Enables the INT0 external interrupt.

Infinite Loop

Section	Register Access	Function
Infinite Loop	(No registers accessed in loop)	The loop is empty ; the entire system runs on interrupts .

Start

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Initialize Ports: PC0/PC1 inputs, PB1/PB2 outputs, PD2 input with pullup

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V

Initialize ADC: Enable, /128 prescaler, Vcc ref

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V

Initialize Timer1: Fast PWM mode 14, /8 prescaler, ICR1=39999, OCR1A/B=3000, enable OVF interrupt

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V

Initialize INT0: Falling edge, enable

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V

Enable global interrupts (sei)

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V

Infinite Loop (empty)

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+-- Timer1 OVF ISR --> Set ADMUX channel 0, start/poll/read ADC, update OCR1A --> Set ADMUX channel 1, start/poll/read ADC, update OCR1B

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+- INT0 ISR --> Set on_ticks1/2=3000, update OCR1A/B